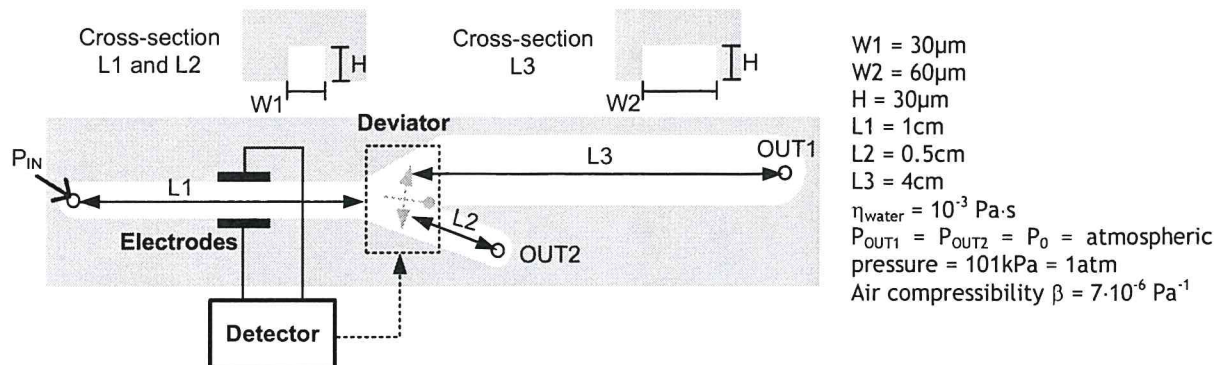


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 Appello 22.07.2014

*Test duration: 2 hours - No support material (textbooks, laptops, smartphones, notes) admitted.
 All the answers must be motivated. Beyond the final numerical results, all the expressions must be provided.*

Exercise 1

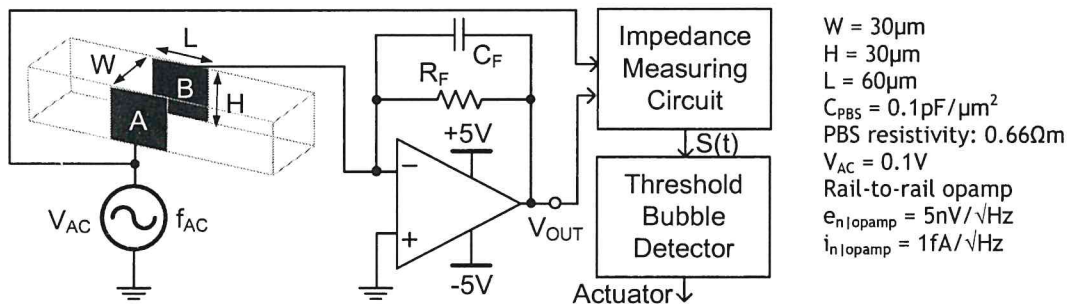
Consider the following microfluidic system for removing air bubbles from the sample: the detector senses the presence of a bubble and drives the actuator to deviate it to OUT2 while the de-bubbled liquid exits from OUT1.



- 16/05 {
- Determine P_{IN} in order to have a fluid speed of $1 \mu\text{m/ms}$ in the electrodes section (no deviator).
 - Determine P_{IN} as in case (a) but for a blood sample instead of water/PBS.
 - Determine the time required to fill the whole channel starting from completely empty condition [P_{IN} as in (a)].
 - If the outlet OUT2 is closed and the section L2 completely filled with air, what is the associated time constant?
 - Without the deviator, what is the efficiency of the device (i.e. the relative amount of lost sample at OUT2)? Propose a solution to improve the efficiency.
 - Propose an implementation of the deviating actuator.
- 22/05 }

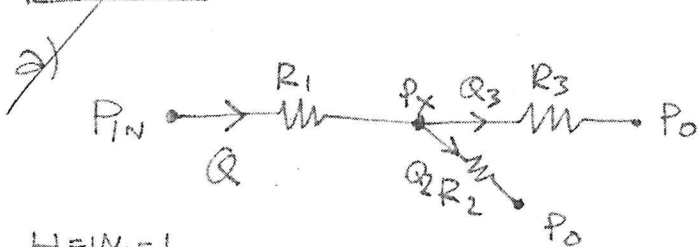
Exercise 2

Now focus on the circuit of the bubble detector of Ex. 1:



- class {
- If the channel is filled with PBS (neglecting all parasitic paths and fringing fields) determine the detector equivalent impedance (between electrodes A and B) and draw it both in the Bode Plot (magnitude and phase) and in the Cole-Cole Plot formats.
 - Draw the detailed scheme of the lock-in impedance measuring block (choosing which parts are required in this case of bubble detection).
 - Determine the value of a suitable sensing frequency f_{AC} , considering also a stray bridge capacitance C_S between A and B of 1 pF .
 - Considering the minimum C_F , determine the value of R_F and plot $V_{OUT}(t)$ in the absence of bubbles, at f_{AC} of (c).
 - Draw the signal $S(t)$ when a bubble of volume equal to twice the sensing volume passes through the sensor at $1 \mu\text{m/ms}$ speed, quoting both amplitude [V] and time [sec] axis's. Consequently choose the proper value of bandwidth of the lock-in filter and the value of the bubble detection threshold.
 - If $R_F = 100 \text{ k}\Omega$, $C_F = 0.5 \text{ pF}$, $C_S = 1 \text{ pF}$ determine and plot the input noise current spectral density (neglect the V_{AC} generator noise and replace C_{DL} with a short circuit).
 - If the rms noise of the whole detection system were 20 ppm , what would be the minimum volume of a detectable air bubble (suggestion: consider the volume fraction).
 - Is it possible to distinguish between an air bubble and a biological cell with the same size? (motivate the answer).

EX. 1



$$V = 1 \mu\text{m/ms}$$



$$Q = V \cdot H \cdot W_1$$

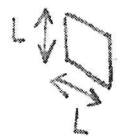
$$= \frac{1 \mu\text{m} \cdot 30 \mu\text{m} \cdot 30 \mu\text{m}}{1 \text{ms}}$$

$$= 0.9 \cdot 10^{-12} \text{ m}^3/\text{s}$$

$$(54 \text{ nL/min})$$

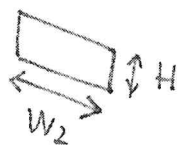
VERY LOW FLOW RATE!

$$H = W_1 = L$$



$$R_{eq} = \frac{2 \text{ Area}}{\text{perimeter}} = \frac{2 L^2}{4 L} = \frac{L}{2} = 15 \mu\text{m}$$

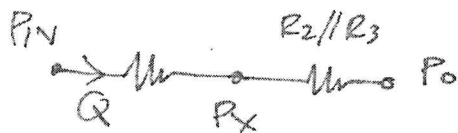
$$R_1 = \frac{8 \eta \cdot L_1}{\pi R_{eq}^4} = \frac{8 \cdot 10^{-3} \cdot 1 \text{ cm}}{\pi (15 \mu\text{m})^4} = 5 \cdot 10^{14} \left[\frac{\text{Pa} \cdot \text{s}}{\text{m}^3} \right]$$



$$\frac{W_2}{H} > 1.4 \Rightarrow$$

$$R_3 = \frac{12 \eta L_3}{(1 - 0.63 \frac{H}{W_2}) H^3 \cdot W_2} = 4.3 \cdot 10^{14} \left[\frac{\text{Pa} \cdot \text{s}}{\text{m}^3} \right]$$

$$R_2 = \frac{R_1}{2} = 2.5 \cdot 10^{14} \left[\frac{\text{Pa} \cdot \text{s}}{\text{m}^3} \right]$$



$$R_2 // R_3 = \frac{2.5 \cdot 4.3}{(2.5 + 4.3)} \cdot 10^{14} = 1.6 \cdot 10^{14}$$

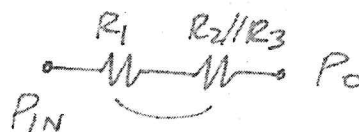
$$P_X = P_O + R_2 // R_3 \cdot Q = 101 \text{ kPa} + 0.9 \cdot 10^{-12} \frac{\text{m}^3}{\text{s}} \cdot 1.6 \cdot 10^{14} \frac{\text{Pa} \cdot \text{s}}{\text{m}^3} = 101.14 \text{ kPa}$$

144 Pa

$$P_{IN} = P_X + Q \cdot R_1 = 101.14 \text{ kPa} + 0.9 \cdot 10^{-12} \frac{\text{m}^3}{\text{s}} \cdot 5 \cdot 10^{14} = 101.59 \text{ kPa}$$

450 Pa

$$\eta_{\text{BLOOD}} = 4 \cdot \eta_{\text{WATER}} \rightarrow$$



$$R_{TOT} = R_1 + R_2 // R_3$$

$$R|_{\text{BLOOD}} = 4 \cdot R_{TOT} = 4 (R_1 + R_2 // R_3)$$

$$P_{IN} = P_O + Q \cdot 4 (R_1 + R_2 // R_3) = 103.37 \text{ kPa}$$

2.37 kPa

(No pressure problems)

$$Q_3 = \frac{P_x - P_0}{R_3} = \frac{144 \text{ Pa}}{4.3 \cdot 10^{14} \left[\frac{\text{Pa} \cdot \text{s}}{\text{m}^3} \right]} = 0.33 \cdot 10^{-12} \text{ m}^3/\text{s}$$

$$Q_2 = \frac{P_x - P_0}{R_2} = \frac{144 \text{ Pa}}{2.5 \cdot 10^{14}} = 0.57 \cdot 10^{-12} \text{ m}^3/\text{s}$$

$$(Q_2 + Q_3 = Q = 0.9 \cdot 10^{-12} \text{ OK})$$

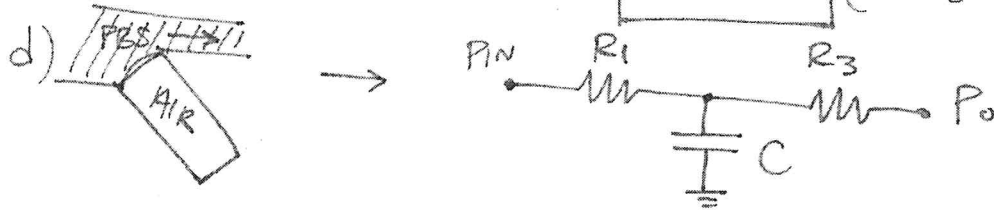
Approx: $t_1 = \frac{L_1 \cdot W_1 \cdot H}{Q} = \frac{1 \text{ cm} \cdot 30 \mu\text{m} \cdot 30 \mu\text{m}}{Q} = 10 \text{ sec}$

tot:

$$t_1 + \max(t_2, t_3) \quad t_2 = \frac{0.5 \text{ cm} \cdot 30 \mu\text{m} \cdot 30 \mu\text{m}}{Q_2} = 7.9 \text{ sec}$$

$$t_3 = \frac{4 \text{ cm} \cdot 30 \mu\text{m} \cdot 60 \mu\text{m}}{Q_3} = 218 \text{ sec}$$

$$t_{\text{TOT}} = 10 \text{ sec} + 218 \text{ sec} = \boxed{228 \text{ sec}} \quad (3.8 \text{ minutes})$$



$$\beta = \frac{1}{V_{\text{ol}}} \cdot \frac{\Delta V_{\text{ol}}}{\Delta P} \Rightarrow C = \frac{\Delta V_{\text{ol}}}{\Delta P} = \beta \cdot V_{\text{ol}}$$

$$V_{\text{ol}}(L_2) = L_2 \cdot W_1 \cdot H = 0.5 \text{ cm} \cdot 30 \mu\text{m} \cdot 30 \mu\text{m} = 4.5 \cdot 10^{-12} \text{ m}^3$$

$$C = 7 \cdot 10^{-6} \frac{1}{\text{Pa}} \cdot 4.5 \cdot 10^{-12} \text{ m}^3 = 31.5 \cdot 10^{-18} \frac{\text{m}^3}{\text{Pa}}$$

$$\tau = (R_1 // R_3) \cdot C = \frac{5 \cdot 4.3}{(5 + 4.3)} \cdot 10^{14} \frac{\text{Pa} \cdot \text{s}}{\text{m}^3} \cdot 31.5 \cdot 10^{-18} \frac{\text{m}^3}{\text{Pa}} =$$

$$= 2.3 \cdot 10^{14} \cdot 31.5 \cdot 10^{-18} = \boxed{7.2 \text{ ms}}$$

$$e) \quad \eta = \frac{Q_3}{Q} = \frac{0.33 \cdot 10^{-12}}{0.9 \cdot 10^{-12}} = \underline{36.6 \%}$$

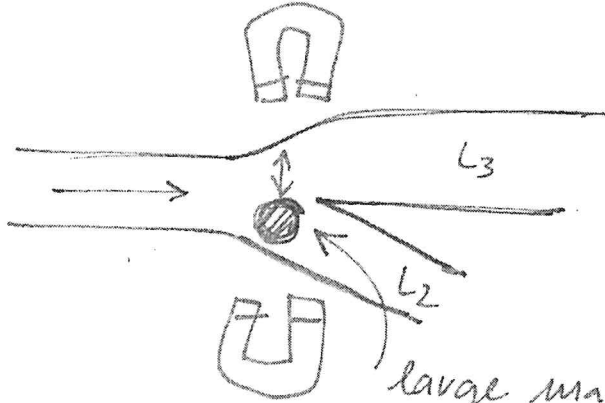
SOLUTION:

(increase L_2 to increase R_2
and thus reduce Q_2)

low efficiency

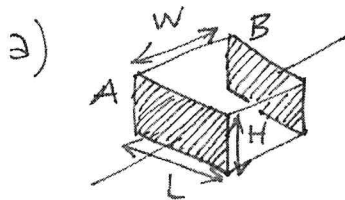
↓
need for the actuator

f)



large magnetic bead normally
closing L_2 driven by external electromagnet.

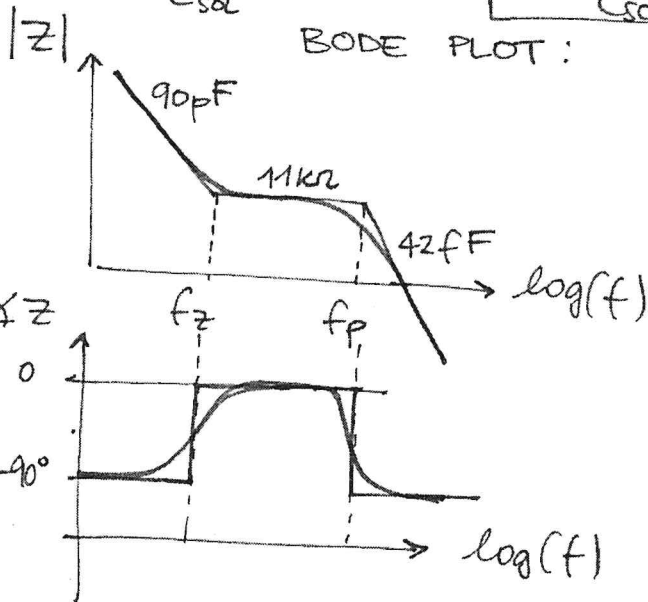
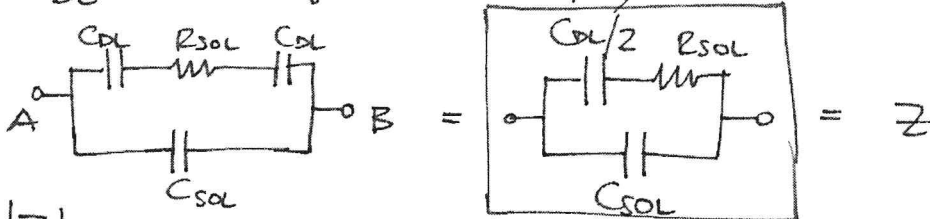
EX. 2



$$R_{sol} = \rho \cdot \frac{W}{LH} = 0.66 \Omega \cdot \text{m} \cdot \frac{30 \mu\text{m}}{30 \mu\text{m} \cdot 60 \mu\text{m}} = 11 \text{ k}\Omega$$

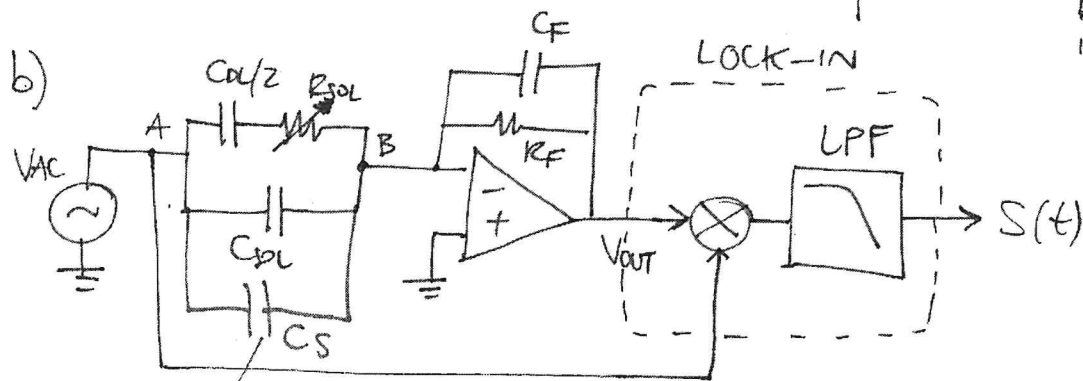
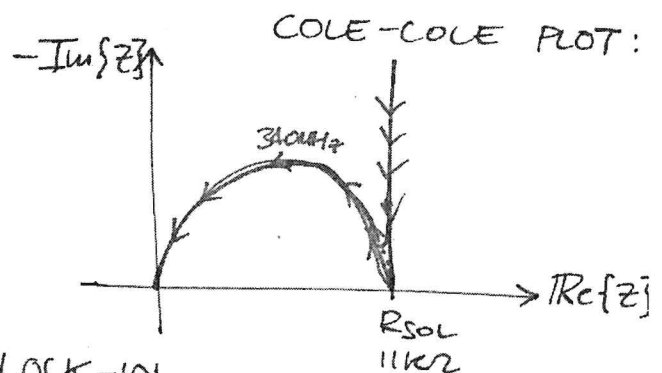
$$C_{sol} = \epsilon_0 \cdot \epsilon_{PSS} \cdot \frac{LH}{W} = \epsilon_0 \cdot 80 \cdot \frac{30 \mu\text{m} \cdot 60 \mu\text{m}}{30 \mu\text{m}} = 42.5 \text{ fF}$$

$$C_{DL} = \text{Area} \cdot C_0 = L \cdot H \cdot 0.1 \text{ pF}/\mu\text{m}^2 = 60 \cdot 30 \cdot 0.1 \text{ pF} = 180 \text{ pF}$$

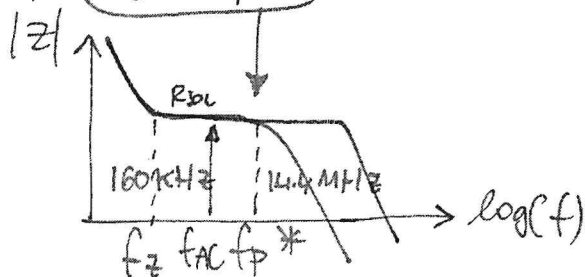


$$f_z = \frac{1}{2\pi R_{sol} \cdot C_{DL}/2} = 160 \text{ kHz}$$

$$f_p = \frac{1}{2\pi R_{sol} \cdot C_{sol}} = 340 \text{ MHz}$$



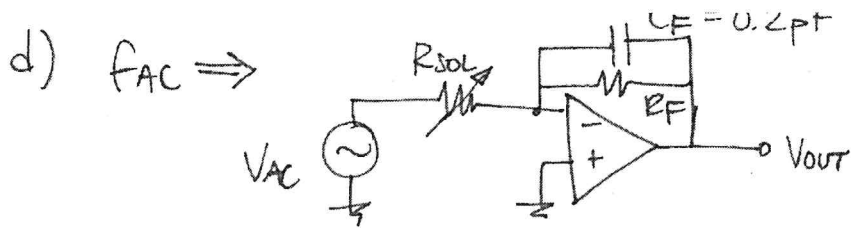
c) $C_S = 1 \text{ pF}$ → Reduces $f_p^* = \frac{1}{2\pi R_{in} \cdot C_S} = 14.4 \text{ MHz}$



$$f_z < f_{AC} < f_p^*$$

$$160 \text{ kHz} < f_{AC} < 14.4 \text{ MHz}$$

$$f_{AC} = 10 \cdot f_z = 1.6 \text{ MHz}$$



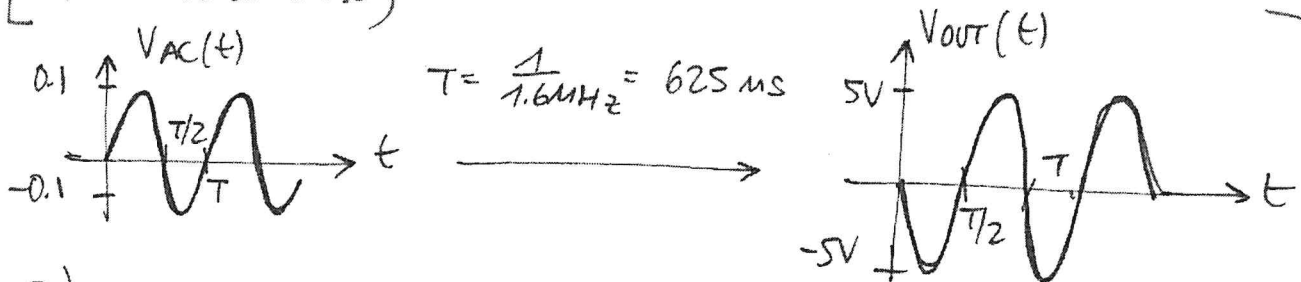
$$V_{OUT} = \frac{V_{AC}}{R_{SOL}} \cdot R_F \Rightarrow V_{OUT}|_{max} = 5V = \frac{0.1V}{11k\Omega} \cdot R_F$$

$$R_F = \frac{5V}{0.1V} \cdot 11k\Omega = \boxed{550k\Omega} \quad \text{full swing (no margin that might be taken...)}$$

↓ CHECK:

$$f_{BW}|_{opamp} \sim \frac{1}{2\pi R_F C_F} = \frac{1}{2\pi \cdot 550k\Omega \cdot 0.2pF} \approx 1.4 MHz \sim f_{AC}$$

(another reason to reduce $R_F \leftarrow$ the pole @ f_{AC} in a real case) NO GOOD!



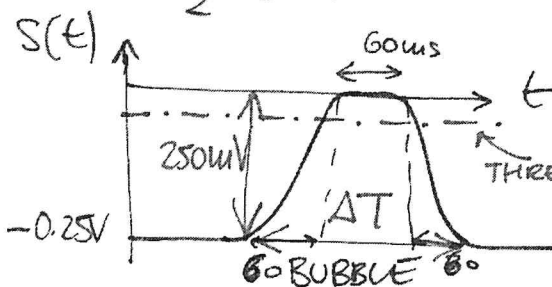
e)

$$S(t) = - \underbrace{\frac{V_{AC}}{R_{SOL}} \cdot R_F}_{V_{OUT}} \cdot V_{AC} \cdot \sin(2\pi f_{AC} t) \cdot \sin(2\pi f_{AC} t)$$

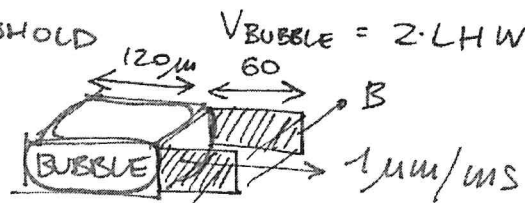
$$= - V_{AC}^2 \cdot \frac{R_F}{R_{SOL}} \cdot \frac{1}{2} \left[\cos(0) - \cos(2\pi \cdot 2 f_{AC} t) \right]$$

LPF

$$= - \frac{1}{2} \cdot (0.1V)^2 \cdot 50 = -0.25V$$



$S=0$ when air fills all the channel

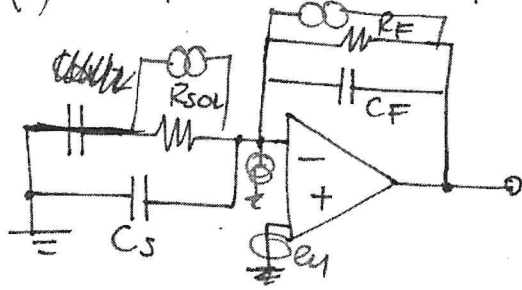


$$\Delta T = \frac{120\mu m}{1\mu m/ms} = 120ms$$

$$BW_{LPF} \sim \frac{1}{\Delta T} \sim \frac{1}{120ms} \sim 8.3Hz$$

$$\tau = \frac{1}{2\pi \cdot 8.3Hz} \approx 1.9\mu s$$

f) $R_F = 100 \text{ k}\Omega$ $C_F = 0.5 \text{ pF}$ $C_S = 1 \text{ pF}$



Noise terms:

$$-\sqrt{\frac{4KT}{R_F}} = 0.4 \text{ pA}/\sqrt{\text{Hz}}$$

$$-\sqrt{\frac{4KT}{R_{\text{SOL}}}} = 1.2 \text{ pA}/\sqrt{\text{Hz}}$$

$$- 2\pi f C_{\text{TOT}} \cdot e_n$$

$$(C_F + C_S)$$

$$- \frac{e_n}{R_F} = \frac{5 \text{ nV}/\sqrt{\text{Hz}}}{100 \text{ k}\Omega} \text{ negligible}$$

$$- \frac{e_n}{R_{\text{SOL}}} = \frac{5 \text{ nV}/\sqrt{\text{Hz}}}{11 \text{ k}\Omega} = 0.45 \text{ pA}/\sqrt{\text{Hz}}$$

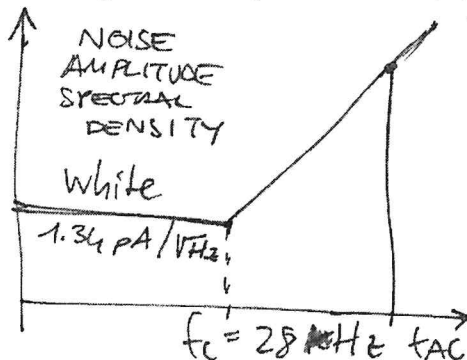
TOT WHITE TERMS:

$$\sqrt{0.4^2 + 1.2^2 + 0.45^2} = 1.34 \text{ pA}/\sqrt{\text{Hz}}$$

CORNER:

$$2\pi f_c \cdot 1.5 \text{ pF} \cdot 5 \text{ nV}/\sqrt{\text{Hz}} = 1.34 \text{ pA}/\sqrt{\text{Hz}}$$

$$f_c = \frac{1.34 \text{ pA}/\sqrt{\text{Hz}}}{(2\pi \cdot 1.5 \text{ pF} \cdot 5 \text{ nV}/\sqrt{\text{Hz}})} = 28.4 \text{ kHz}$$



@ f_{AC} , e_n dominates
 (f_{AC} should be reduced!!)
 look for:
 max SNR(f)...

g) $\sigma_{\text{NOISE}} = 20 \text{ ppm}$

$$\Delta V_{ol}|_{\text{min}} \cong V_{ol} \cdot 20 \text{ ppm}$$

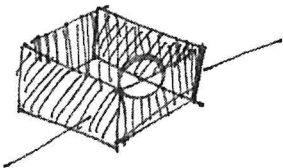
$$= W \cdot H \cdot L \cdot 20 \text{ ppm} =$$

$$= 60 \cdot 30 \cdot 30 \cdot (\mu\text{m})^3 \cdot 20 \text{ ppm}$$

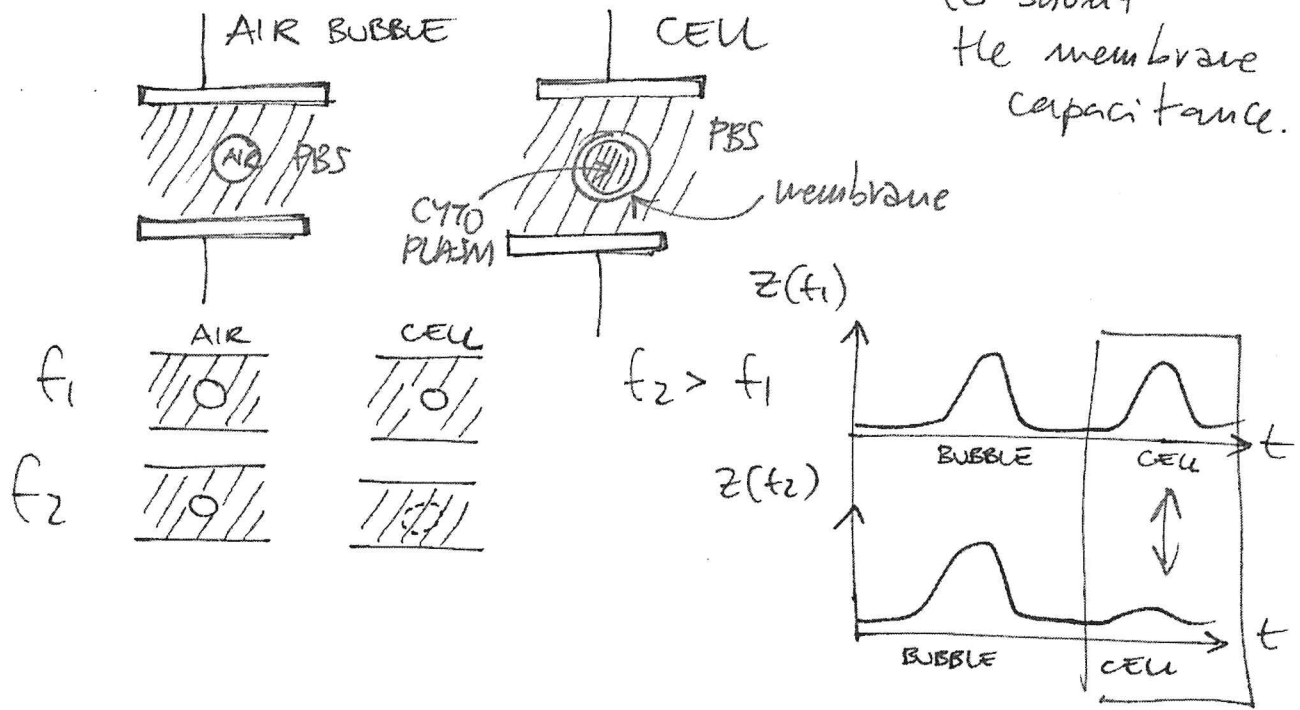
$$= 1 \mu\text{m}^3$$

$$\frac{4}{3}\pi r^3 = 1 \mu\text{m}^3 \Rightarrow$$

$$r = 0.62 \mu\text{m} \text{ MIN. DIAMETER}$$



h) YES, it is possible: USING more frequencies to shunt the membrane capacitance.

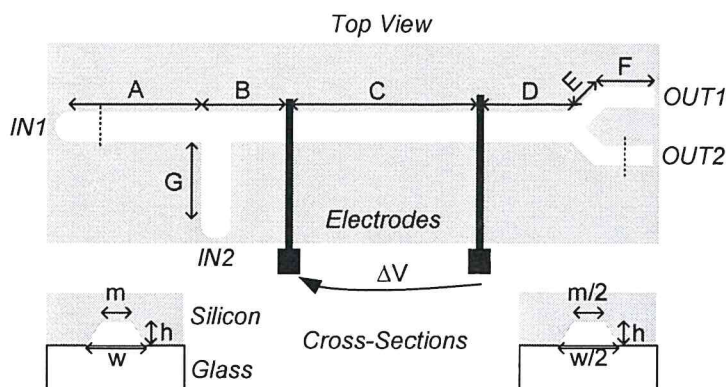


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Exercise 1

Consider the following microfluidic system (with 2 inputs and 2 outputs) realized by etching a channel with trapezoidal cross-section in Silicon. Sections A-B-G represent a T-junction, section C is between two electrodes fabricated on the glass substrate, D-E-F represent the bifurcated output section. Adopt proper approximations.

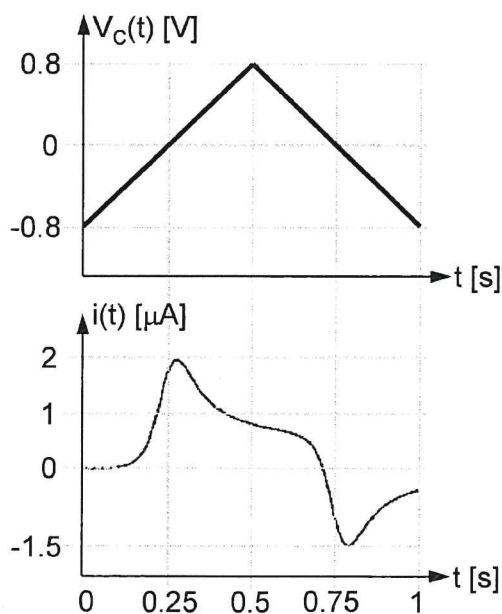


$A = 1\text{mm}$
 $B = 0.5\text{mm}$
 $C = 1\text{mm}$
 $D = 0.2\text{mm}$
 $E = 0.2\text{mm}$
 $F = 0.5\text{mm}$
 $G = 0.2\text{mm}$
 $w = 50\mu\text{m}$
 $m = 30\mu\text{m}$
 $h = 30\mu\text{m}$
 $\eta_{\text{water}} = 10^{-3}\text{ Pa}\cdot\text{s}$
 $\xi = 50\text{mV}$ (zeta potential)
 $P_0 = \text{atmospheric pressure} = 101\text{kPa} = 1\text{atm}$

- If $P_{\text{IN1}} = 109\text{kPa}$, $P_{\text{OUT1}} = P_{\text{OUT2}} = P_0$, find P_{IN2} in order to achieve a flow velocity of 1mm/min in section C. Determine also the flow velocities in all the sections.
- How does the velocity in section C changes if output 2 gets clogged?
- If water enters into IN1 and PBS enters into IN2, find and draw the liquid separation (cross-section) in section C.
- Considering a PBS solution filling the channels, find the voltage difference ΔV to be applied to the electrodes in order to obtain in section C the same fluid velocity of point (a) by means of electro-osmotic flow.
- Now consider a spherical particle of diameter $10\mu\text{m}$ and charge $Q = 1000$ electrons in deionized water, find the particle direction and velocity if $\Delta V = +10\text{V}$.

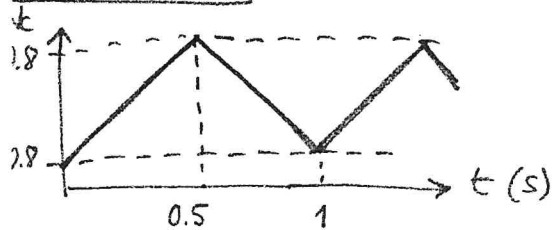
Exercise 2

Considering the following waveforms corresponding to one cycle of a *cyclic voltammetry* (CV) three-electrode measurement with a FeCN redox species in concentration $10\mu\text{M}$: $V_c(t)$ is the voltage applied to the electrochemical cell (with respect to Ag/AgCl RE) and $i(t)$ is the current sensed at the working electrode (WE):



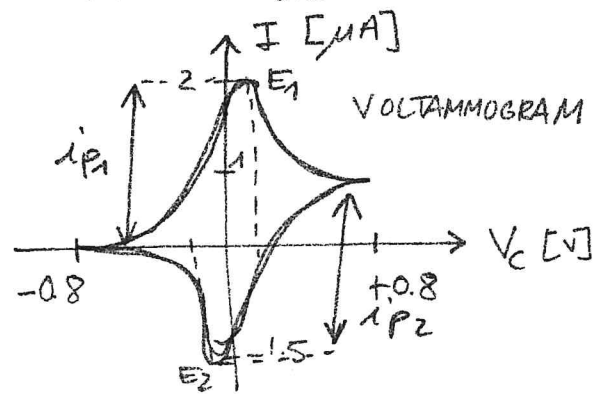
- Determine the *scan-rate* of this CV.
- Plot the *voltammogram* (with quantitative axes).
- Determine the values of the reduction (E_1) and oxidation (E_2) potentials (with respect to the Ag/AgCl reference electrode). Is the reaction reversible?
- Given this *scan rate*, determine the suitable bandwidth of the potentiostat.
- Considering a max. tolerable error of 10% of $\Delta E = E_1 - E_2$ in ideally reversible conditions, what is the accuracy required for the voltage generator source (V_c)?
- Draw the scheme of the current reader connected to the WE and choose the value of the feedback resistor of the transimpedance amplifiers considering a dual supply voltage of $\pm 5\text{V}$.
- If the concentration of the FeCN redox species is reduced to $1\mu\text{M}$, how does the voltammogram change?
- If a small-signal impedance measurement is performed on this electrochemical system, find the DC bias potential V_c at which the lowest charge transfer resistance (R_{CT} in this Faradaic interface) is measured.
- If the electrodes for this CV measurement are to be inserted into a miniaturized biochip and thus fabricated in a standard CMOS microelectronic process, what are the issues from the materials point of view.

EXERCISE 2



$$a) \quad SR = \frac{dV}{dt} = \frac{0.8 + 0.8}{0.5} = 3.2 \text{ V/s}$$

b)



c) from the voltammogram, :

$$E_1 \sim 0.12 \text{ V}$$

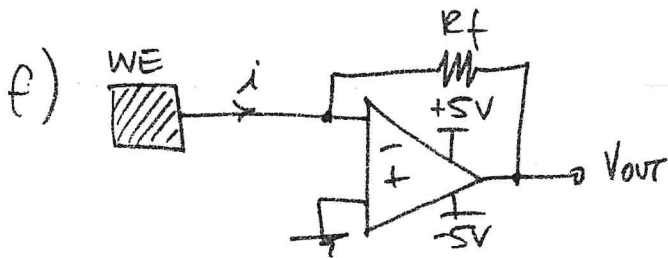
$$E_2 \sim -0.12 \text{ V}$$

$\Delta E = E_1 - E_2 \sim 240 \text{ mV} > 59 \text{ mV}$ (ideal separation), but since $i_{p1} \cong i_{p2}$ the reaction is reversible.
($\sim 2 \mu\text{A}$)

$$d) \quad BW = 40 \cdot SR \Rightarrow 40 \cdot 3.2 = 128 \text{ Hz min. bandwidth}$$

$$e) \quad \Delta E_{\text{ideal}} = 59 \text{ mV} \quad E_{\text{max}} = 10\% \cdot \Delta E = 5.9 \text{ mV}$$

Accuracy better than 5.9 mV

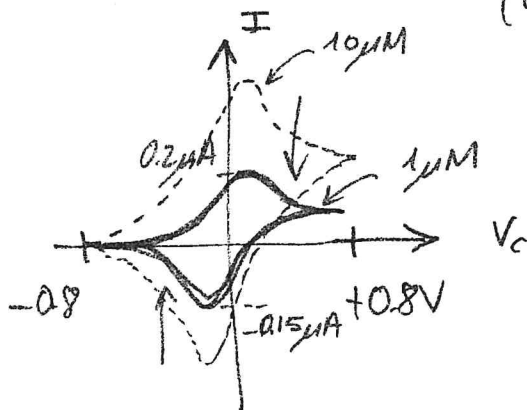


$$i_{\text{max}} = 2 \mu\text{A}$$

$$i_{\text{min}} = -1.5 \mu\text{A}$$

$$Rf = \frac{5 \text{ V}}{2 \mu\text{A}} = 2.5 \text{ M}\Omega$$

g) Linear scaling of the current by a factor 10:
(with the concentration)



NEXT PAGE 5

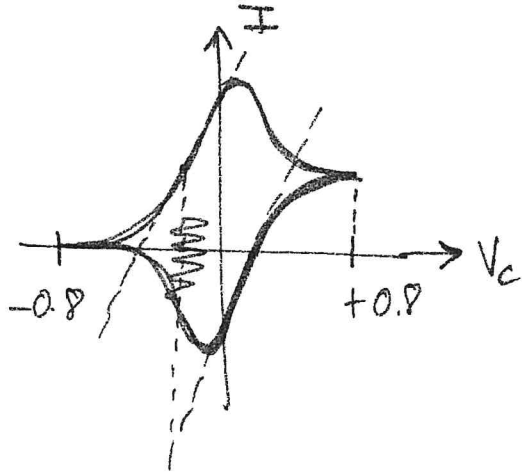
$$h) R_{CT} = \left(\frac{\partial i}{\partial V_c} \right)^{-1}$$

$R_{CT} \text{ min} \rightarrow$ highest slope of the voltammogram

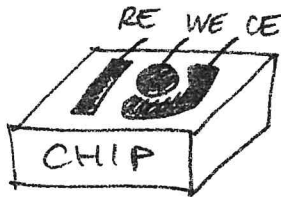
from the plot

$\frac{\partial i}{\partial V_c} / \text{max}$ in

$V_c = -0.1V$
and
 $V_c \sim 0V$



i) CMOS: ELECTRODES ISSUES



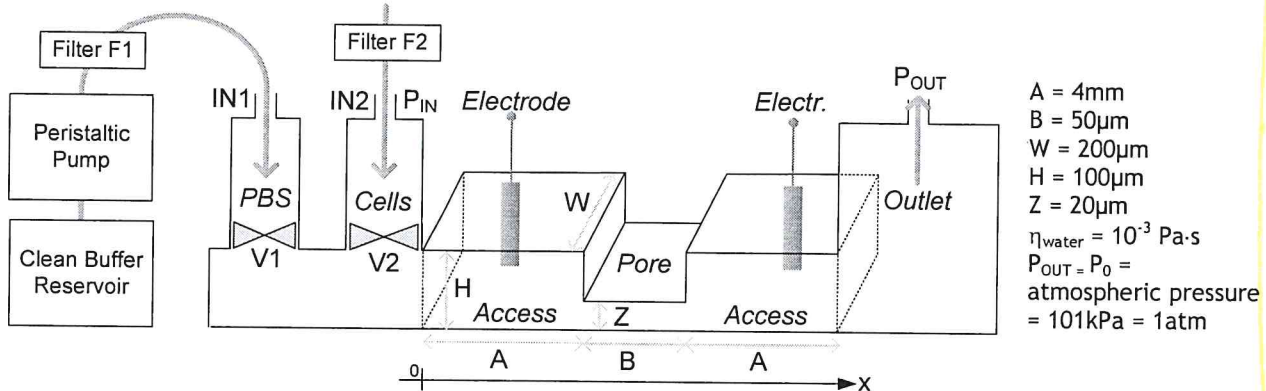
from Al $\rightarrow$ WE $\rightarrow$ GOLD
 $\rightarrow$ RE $\rightarrow$ Ag/AgCl

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Exercise 1

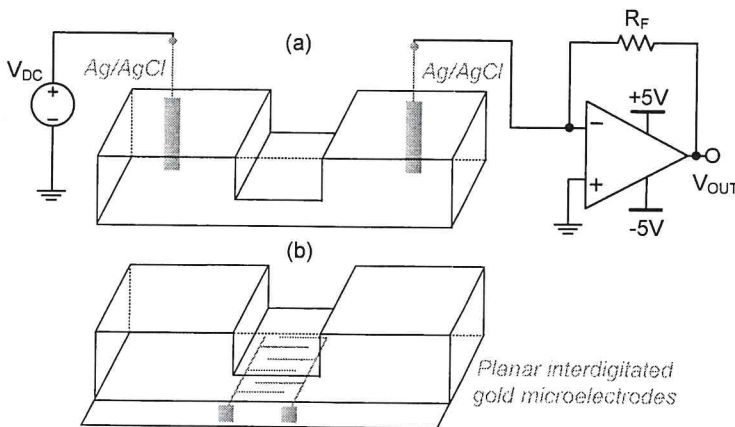
Consider the following microfluidic system implementing a Coulter counter for cell counting and sizing by means of a micro channel (Pore) of length B and of rectangular cross-section Z by W (access sections are required to allow the space for electrodes). Inlet 1 (IN1, controlled by valve V1) is used to flow PBS cleaning buffer, while inlet 2 (IN2, controlled by valve V2) is used to inject the sample with cells suspended in PBS as well.



- Focusing only on the A-B-A sections, if $P_{\text{IN}} = 2\text{atm}$, calculate the resulting flow rate.
- Draw the scheme of a standard **off-chip** peristaltic pump used to pump the PBS buffer (for system cleaning) and discuss a possible implementation of an **on-chip** peristaltic pump.
- Considering the framework of electrical analogies of fluidic circuits, what is the electrical equivalent of fluid velocity?
- Calculate and plot the cell velocity along the x axis (A-B-A) assuming an input flow rate $Q_{\text{IN}} = 10\mu\text{l/s}$.
- Determine the **cut-off dimension** (i.e. the min. size of stopped impurities) of microfiltration filters F1 and F2.
- Draw and discuss an implementation of valves V1 and V2. Should they be active or passive valves?
- Assuming that inlet IN1 is empty (filled with air instead of PBS) and V1 is closed by a flexible membrane of area $200\mu\text{m} \times 200\mu\text{m}$ and elastic constant $k = 0.2\text{N/m}$, find the associated time constant τ and the maximum rotation speed (in rpm) of the motor of a peristaltic pump (the same described in point b) driving IN2 and producing a periodic pulsation that is not attenuated by the channel intrinsic τ .

Exercise 2

Now focus on the detection circuit for the above system, assuming that the pore resistance variation is proportional to the volume fraction occupied by the cell. Compare detection approach (a) versus (b).



$R_F = 500\text{k}\Omega$, $V_{\text{DC}} = 1\text{V}$
 $\sigma_{\text{PBS}} = 1.5\text{S/m}$

For scheme (a) - Ag/AgCl electrodes:

- Draw the impedance model between the electrodes when the pore is filled with PBS. What is the voltage drop across the pore resistance? How can it be improved?
- Find V_{OUT} in DC in the absence of cells.
- Given the speed found in point 1.d, what is the min. bandwidth of the amplifier? Is it compatible with this detection scheme?
- With the same speed, plot $V_{\text{OUT}}(t)$ due to the passage of a cell of $10\mu\text{m}$ diameter.
- If the global resolution of the detector is 10ppm, what are the min. and max. diameters of detectable cells?

For scheme (b) - bottom planar Au interdigitated microelectrodes:

- Draw the detection circuit and the small-signal impedance equivalent model. How does it differ from case (a)?
- What is the optimal size of the gap between the interdigitated electrodes?
- What is the alignment accuracy required to align the bottom glass with patterned gold microelectrodes with the top channel?

EX. 1

1/5

a) $P_{in} \xrightarrow{R_A} \xrightarrow{R_B} \xrightarrow{R_A} P_{out} = P_0 \quad R_{TOT} = 2R_A + R_B$

$$R_A = \frac{12 \eta \cdot A}{(1 - 0.63 \frac{h}{w}) h^3 w} = \left[\frac{W}{H} = 2 > 1.4 \right] = 350.4 \cdot 10^9 \left[\frac{Pa \cdot s}{m^3} \right]$$

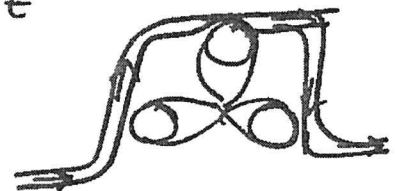
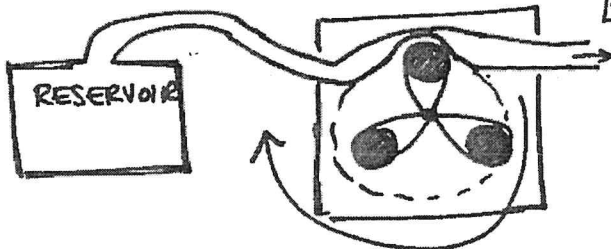
$$R_B = \frac{12 \eta B}{(1 - 0.63 \frac{z}{w}) z^3 \cdot w} = \left[\frac{W}{z} = 10 > 1.4 \right] = 400.2 \cdot 10^9 \left[\frac{Pa \cdot s}{m^3} \right]$$

$$R_{TOT} = 2 \cdot 350.4 + 400.2 = 1101 \cdot 10^9 \frac{Pa \cdot s}{m^3} = 1.1 \cdot 10^{12} \frac{Pa \cdot s}{m^3}$$

$$Q = \frac{\Delta P}{R_{TOT}} = \frac{P_{in} - P_{out}}{R_{TOT}} = \frac{2 \text{ atm} - 1 \text{ atm}}{R_{TOT}} = \frac{1 \text{ atm}}{R_{TOT}} = 92 \cdot 10^{-9} \frac{m^3}{s}$$

(92 $\mu\text{e/s}$)

b) PERISTALTIC PUMP: $Q \uparrow$



c) FLUID:

$$\text{Velocity} \left[\frac{m}{s} \right] = \frac{\text{Flow rate } Q \left[\frac{m^3}{s} \right]}{\text{cross-section area } w \cdot h \left[m^2 \right]}$$

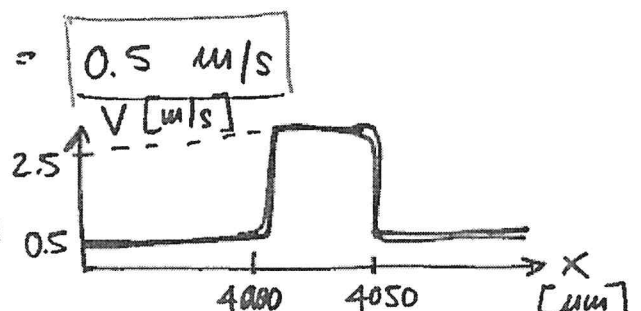
ELECTRICAL EQUIVALENT:

$$\frac{Q}{\text{Area}} \Rightarrow \frac{I}{\text{Area}} \text{ current} = J \left[\frac{A}{m^2} \right] \quad \text{CURRENT DENSITY}$$

d) $Q = 10 \mu\text{e/s} = 10 \cdot 10^{-6} \frac{l}{s} = 10 \cdot 10^{-9} \frac{m^3}{s}$

$$V_A = \frac{10^{-8} \frac{m^3}{s}}{H \cdot W} = \frac{10^{-8}}{100 \cdot 10^{-6} \cdot 200 \cdot 10^{-6}} = \boxed{0.5 \text{ m/s}}$$

$$V_B = \frac{10^{-8}}{z \cdot w} = 5 \cdot V_A = \boxed{2.5 \text{ m/s}}$$

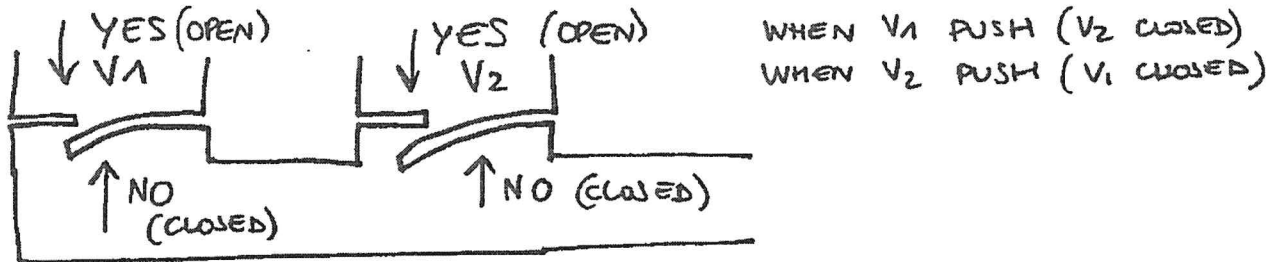


e) $F1 \Rightarrow$ filters PBS $\rightarrow$ THE SMALLEST POSSIBLE
 $\phi \sim$ A FEW NANOMETERS

2/5

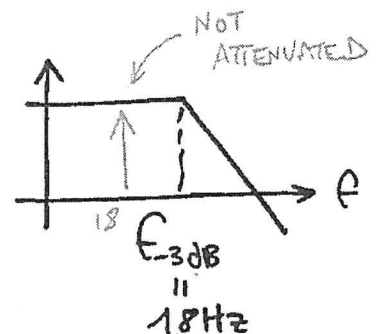
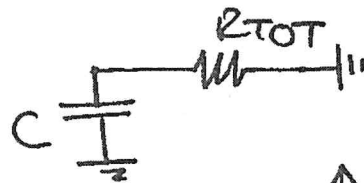
$F2 \Rightarrow$ CELLS MUST PASS $\rightarrow \phi_{min} = z = 20 \mu m$

f) PASSIVE MONODIRECTIONAL VALVES



g) Capacitance = $\frac{A^2}{K}$

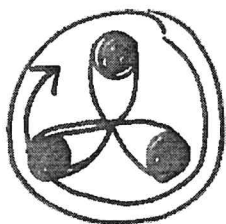
$$= \frac{(200 \mu m \cdot 200 \mu m)^2}{0.2 N/m} = 8 \cdot 10^{-15}$$



$$\tau = C \cdot R_{TOT} = 8 \cdot 10^{-15} \cdot 1.1 \cdot 10^{12} = \boxed{8.8 \mu s}$$

$$f_{-3dB} = \frac{1}{2\pi\tau} = 18 \text{ Hz}$$

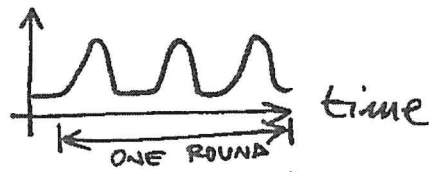
$$f_{MAX} = \frac{f_{-3dB}}{10} \text{ (ONE DECADE MARGIN)} = 1.8 \text{ Hz}$$



PERISTALTIC
PUMP



3 cylinders $\Rightarrow$ 3 pulses/round



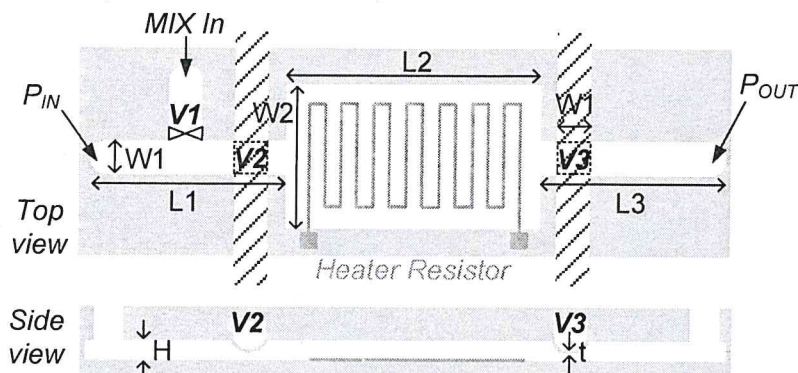
$$\frac{1.8 \text{ pulses/sec}}{3 \text{ pulses/round}} = \omega_{max} = 0.6 \text{ round/s} \cdot \frac{60}{60} = \boxed{36 \text{ rpm}}$$

Politecnico di Milano
Biochip A.A. 2013/2014 - M. Carminati
 Appello 3.09.2014

*Test duration: 2 hours - No support material (textbooks, laptops, smartphones, notes) admitted.
 All the answers must be motivated. Beyond the final numerical results, all the expressions must be provided.*

Exercise 1

Consider the following PDMS microfluidic system for PCR amplification of the input sample in a single chamber whose temperature is controlled by means of a thin-film resistor. Flow is controlled by 3 valves V1, V2 and V3:

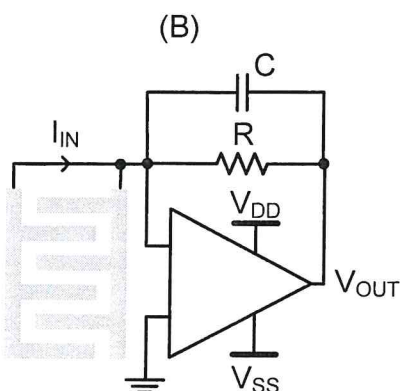
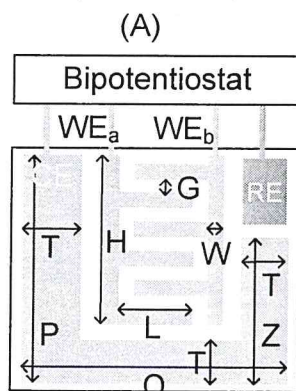


$W1 = 200\mu\text{m}$
 $W2 = 1\text{mm}$
 $L1 = 2\text{cm}$
 $L2 = 1\text{cm}$
 $L3 = 1\text{cm}$
 $H = 40\mu\text{m}$
 $t = 7\mu\text{m}$ (membrane thickness)
 $k = 20\text{N/m}$ (membrane spring constant)
 $\eta_{\text{water}} = 10^{-3}\text{ Pa}\cdot\text{s}$
 $P_{\text{OUT}} = \text{atmospheric pressure} = 101\text{kPa} = 1\text{atm}$

- Given the input pressure $P_{\text{IN}} = 110\text{kPa}$ (generated by an external syringe pump), V1 closed, V2 open and V3 open, and neglecting transients, find the flow speed in the three sections, L1, L2 and L3.
- Briefly describe the operating principles of valves V2 and V3.
- Neglecting the MIX, what is the volume of sample that can be processed and estimate the chamber filling time.
- Considering only V3, find the *hydrodynamic capacitance* associated with the thin square flexible membrane.
- Determine the highest temperature that the liquid in the reaction chamber must reach during the PCR process.
- Draw the temperature temporal sequence $T(\text{time})$ for a complete PCR cycle, describing with a single-line sentence the different phases.
- Propose a system for mixing *on chip* the sample solution and the MIX reagents injected from a separate input.

Exercise 2

Consider the following set of microfabricated planar electrodes, including a pair of interdigitated working electrodes (WE_a and WE_b) with 3 fingers each:

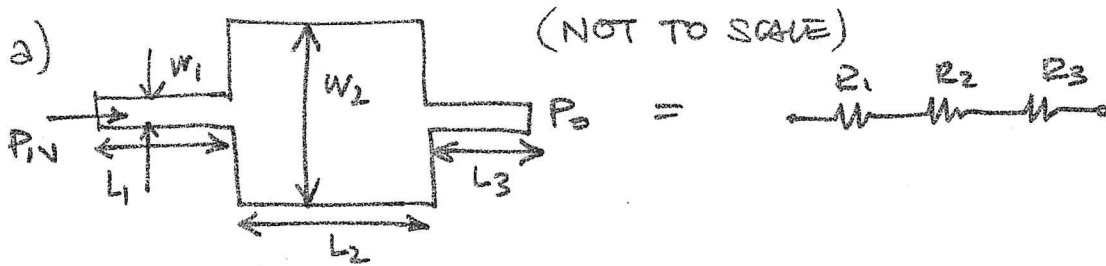


$W = 5\mu\text{m}$
 $G = 5\mu\text{m}$
 $H = 60\mu\text{m}$
 $L = 50\mu\text{m}$
 $T = 30\mu\text{m}$
 $P = 100\mu\text{m}$
 $Q = 100\mu\text{m}$
 $Z = 60\mu\text{m}$
 $C_{\text{DL-PBS}} = 0.1\text{pF}/\mu\text{m}^2$
 $R = 100\text{k}\Omega$

Draw the correct signs of the opamp input terminals

- Draw the detailed scheme of a Bipotentiostat connected to the electrodes shown in (A).
- For the Bipotentiostat of point a), set the biases in the circuit in DC in order to have the potential of $\text{WE}_a = +0.5\text{V}$ (with respect to the solution), and the potential of $\text{WE}_b = -0.3\text{V}$ (with respect to the solution).
 Now consider both working electrodes connected together to a single current amplifier as shown in (B):
- Determine the value of the double layer capacitance C_{DL} connected at the amplifier input when the electrodes are in PBS. Is the C_{DL} of the counter electrode (CE) negligible?
- Considering an ideal opamp and only the thermal noise of the feedback resistor R, determine the value of the bandwidth BW (and consequently of C) in order to have an input current resolution of 100pA RMS.
- Now considering the input equivalent voltage noise source of the opamp $e_n = 3\text{nV}/\sqrt{\text{Hz}}$, determine the value the corner frequency and plot the input-referred noise spectrum.
- Considering a cyclic voltammetry measure in the macroscopic regime, if the area of the working electrodes is reduced by a factor 2, how should the *scan rate* be modified to measure the same input current?
- What is the spatial accuracy of the *mask aligner* necessary to fabricate the electrodes in (A)?
- Propose a scheme to measure the total current flowing from the COUNTER electrode (CE).

EXERCISE 1



$\frac{W}{h} > 1.4 \rightarrow R = \frac{12 \gamma L}{(1 - 0.63 \frac{h}{W}) h^3 \cdot W}$

$$R_3 = R_1/2$$

$$R_1 = \frac{12 \gamma L_1}{(1 - 0.63 \frac{h}{W_1}) h^3 \cdot W_1} = \frac{12 \cdot 10^{-3} \cdot 2 \text{ cm}}{(1 - 0.63 \frac{40}{200}) (40 \cdot 10^{-6})^3 \cdot 200 \cdot 10^{-6}} = 21.4 \cdot 10^{12} \left[\frac{\text{Pa} \cdot \text{s}}{\mu\text{m}^3} \right]$$

$$R_3 = 10.7 \cdot 10^{12} [\text{Pa} \cdot \text{s} / \mu\text{m}^3]$$

$$R_2 = \frac{12 \gamma L_2}{(1 - 0.63 \frac{h}{W_2}) h^3 \cdot W_2} = \frac{12 \cdot 10^{-3} \cdot 1 \text{ cm}}{(1 - 0.63 \frac{40}{1000}) (40 \cdot 10^{-6})^3 \cdot 1000 \cdot 10^{-6}} = 2 \cdot 10^{12} \left[\frac{\text{Pa} \cdot \text{s}}{\mu\text{m}^3} \right]$$

$$R_{\text{TOT}} = R_1 + R_2 + R_3 = 34.1 \cdot 10^{12} [\text{Pa} \cdot \text{s} / \mu\text{m}^3]$$

$Q = \frac{\Delta P}{R} = \frac{110 \text{ kPa} - 101 \text{ kPa}}{34.1 \cdot 10^{12} \left[\frac{\text{Pa} \cdot \text{s}}{\mu\text{m}^3} \right]} =$

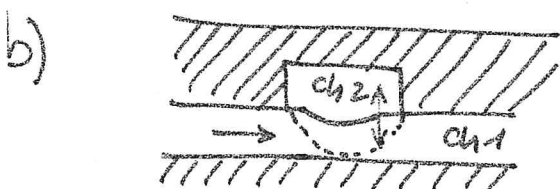
$$= \frac{9 \text{ kPa}}{34.1 \cdot 10^{12} \dots} = \boxed{\frac{264 \cdot 10^{-12} \text{ m}^3/\text{s}}{(15.8 \text{ } \mu\text{e}/\text{min})}}$$

FLOW-RATE

$$(1 \mu\text{m}^3 = 1000 \text{ e})$$

$$V_1 = \frac{Q}{W_1 \cdot H} = \frac{264 \cdot 10^{-12} \frac{\text{m}^3}{\text{s}}}{40 \mu\text{m} \cdot 200 \mu\text{m}} = 33 \text{ mm/s}$$

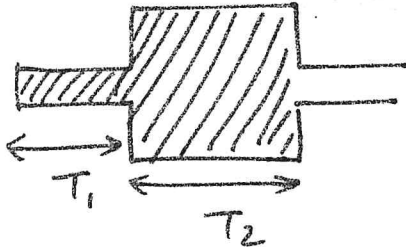
$$V_2 = \frac{Q}{W_2 \cdot H} = \frac{V_1}{5} = 6.6 \text{ mm/s}$$



Pressure in channel 2 controls the flexible membrane that opens/closes channel 1.

c) Volume / PCR-chamber = $W_2 \cdot H \cdot L_2 =$
 $= 40 \mu\text{m} \cdot 2000 \mu\text{m} \cdot 1 \text{ cm} = 400 \cdot 10^{-12} \text{ cm}^3$
 $= (0.4 \mu\text{e})$

Filling time: $T_{\text{TOT}} \sim T_1 + T_2$

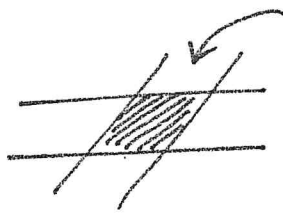


$$T_1 = \frac{L_1}{V_1} = \frac{2 \text{ cm}}{33 \text{ mm/s}} = 0.6 \text{ sec}$$

$$T_2 = \frac{L_2}{V_2} = \frac{1 \text{ cm}}{6.6 \text{ mm/s}} = 1.5 \text{ sec}$$

$$T_{\text{TOT}} = \frac{2.1 \text{ sec}}{2} = 1.05 \text{ sec}$$

d)



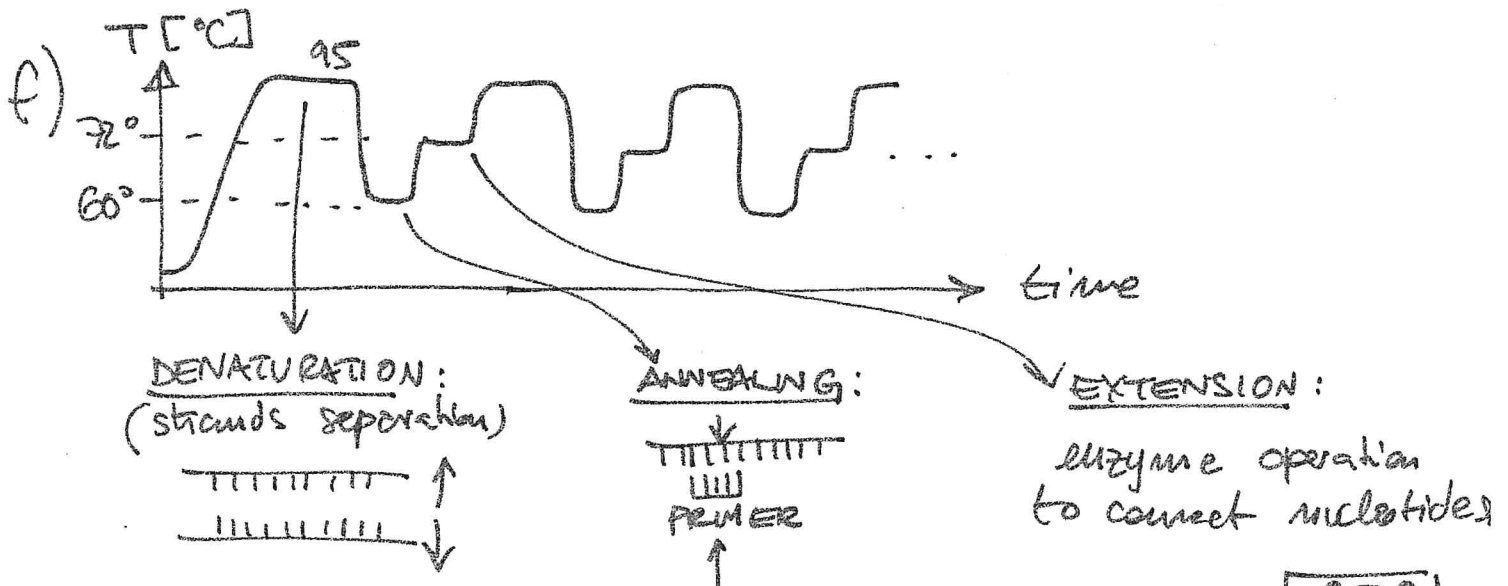
flexible

$$C_H = \frac{A_{\text{mem}}^2}{K} = \frac{(200 \mu\text{m} \cdot 200 \mu\text{m})^2}{20 \text{ N/m}}$$

$$2 \cdot 10^{14} \text{ m}^3/\text{Pa}$$

$$\tau = \underbrace{(R_1 + R_2) // R_3}_{7.3 \cdot 10^{12}} \cdot \underbrace{C_H}_{8 \cdot 10^{-17}} = \underbrace{0.6 \text{ ms}}_{\text{FAST!}}$$

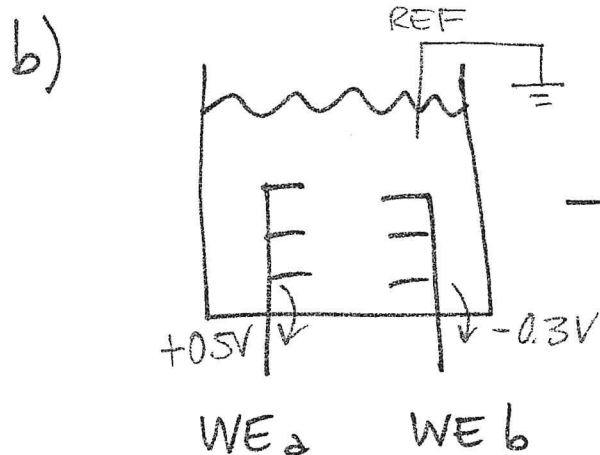
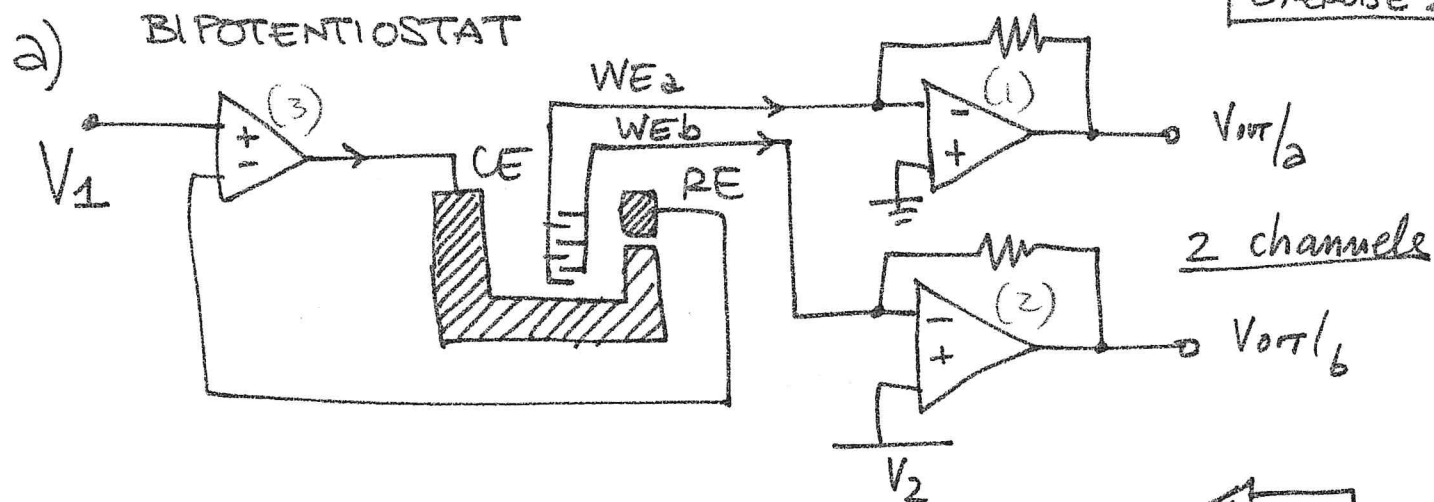
e) $T_{\text{max}} = T_{\text{DNA denaturation}} = 95^\circ\text{C}$



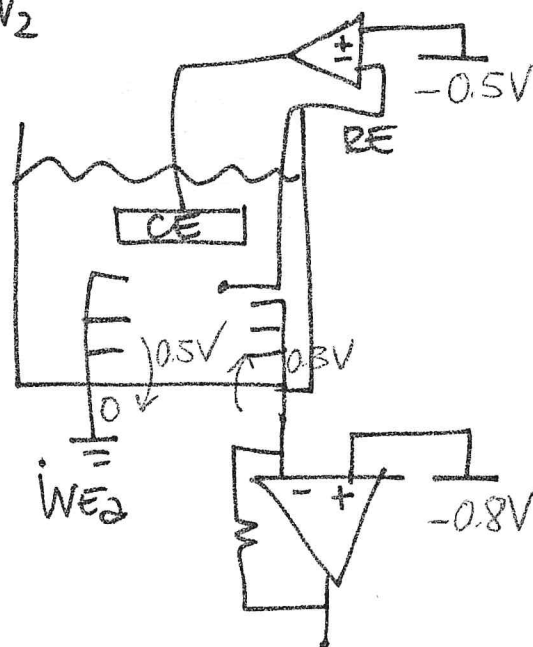
g) MAGNETIC BEADS INSIDE THE CHAMBER (ROTATING MAGNET)



a) BI-POTENTIOSTAT



SHIFT



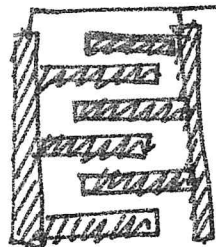
$$\begin{aligned} V_1 &= -0.5V \\ V_2 &= -0.8V \end{aligned}$$

or

$$\begin{aligned} V_1 &= 0V \\ V_2 &= -0.3V \\ (V_{xc} &= +0.5V \text{ VIRTUAL GROUND 1st opamp}) \end{aligned}$$

c) $C_{DL} = \text{AREA} \cdot C_0$ (IN PARALLEL)

AREA :



$$\begin{aligned} A &= 6 \cdot L \cdot W + 2 \cdot H \cdot W = \\ &= W(6L + 2H) = 5(6 \cdot 50 + 2 \cdot 60) \\ &= 2100 \mu m^2 \end{aligned}$$

$$C_{DL}|_{WE} = 2100 \mu m^2 \cdot 0.1 \frac{pF}{\mu m^2} = \underline{210 pF}$$

$$C_{DL}|_{CE} = \text{AREA}_{CE} \cdot C_0 = T \cdot P + 2 \cdot P + (Q - 2T) \cdot T$$

$$= T(P + 2 + Q - 2T) =$$

$$= 30(100 + 60 + 100 - 60) = 6000 \mu m^2$$

3 times $C_{DL}|_{WE}$

d) Thermal noise:

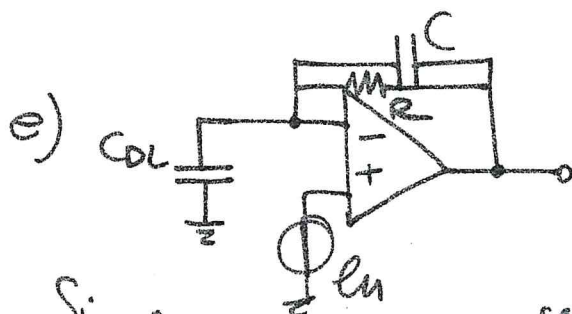
$$\frac{4kT}{R} \rightarrow \sigma_{min} = \sqrt{\frac{4kT}{R} \cdot \frac{\pi}{2} \cdot BW} = 100 \text{ pA RMS}$$

$$\sqrt{\frac{4kT}{9} \cdot \frac{9}{R}} = 0.4 \text{ pA}/\sqrt{\text{Hz}}$$

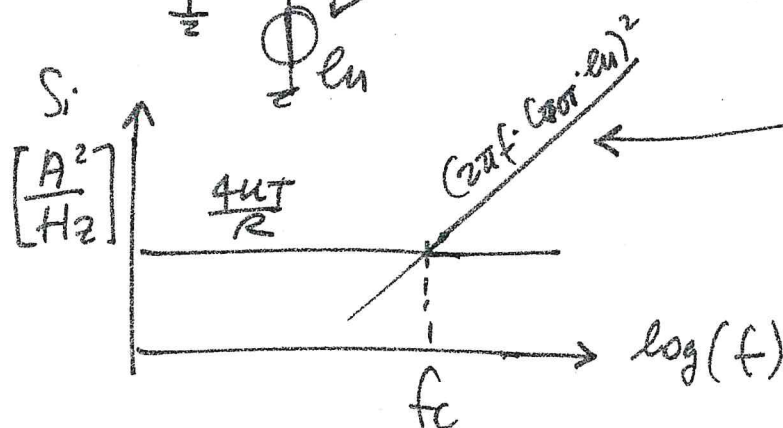
$$0.2 \text{ pA}/\sqrt{\text{Hz}} \cdot \sqrt{\frac{\pi}{2} \cdot BW} = 100 \text{ pA} \Rightarrow BW = \left(\frac{100 \text{ pA}}{0.2 \text{ pA}/\sqrt{\text{Hz}}} \cdot \frac{2}{\pi} \right)^2$$

$$BW = \frac{1}{2\pi RC} \rightarrow C = \frac{1}{2\pi RBW} = 15.7 = 10^4 \text{ kHz} = 25$$

$$= 16 \text{ pF} \cdot H = 64$$



$$S_i / I_{TOT} = \frac{4kT}{R} + (2\pi f \cdot C_{TOT} \cdot e_n)^2$$



$$C_{TOT} = C_{OL} + C = 210 + 16 = 226 \text{ pF}$$

$$\frac{4kT}{R} = (2\pi f \cdot C_{TOT} \cdot e_n)^2 \Rightarrow f_c = \frac{\sqrt{\frac{4kT}{R}}}{2\pi C_{TOT} \cdot e_n} = \frac{0.4 \text{ pA}/\sqrt{\text{Hz}}}{2\pi \cdot 226 \text{ pF} \cdot e_n}$$

$$= \frac{77}{47} \text{ kHz}$$

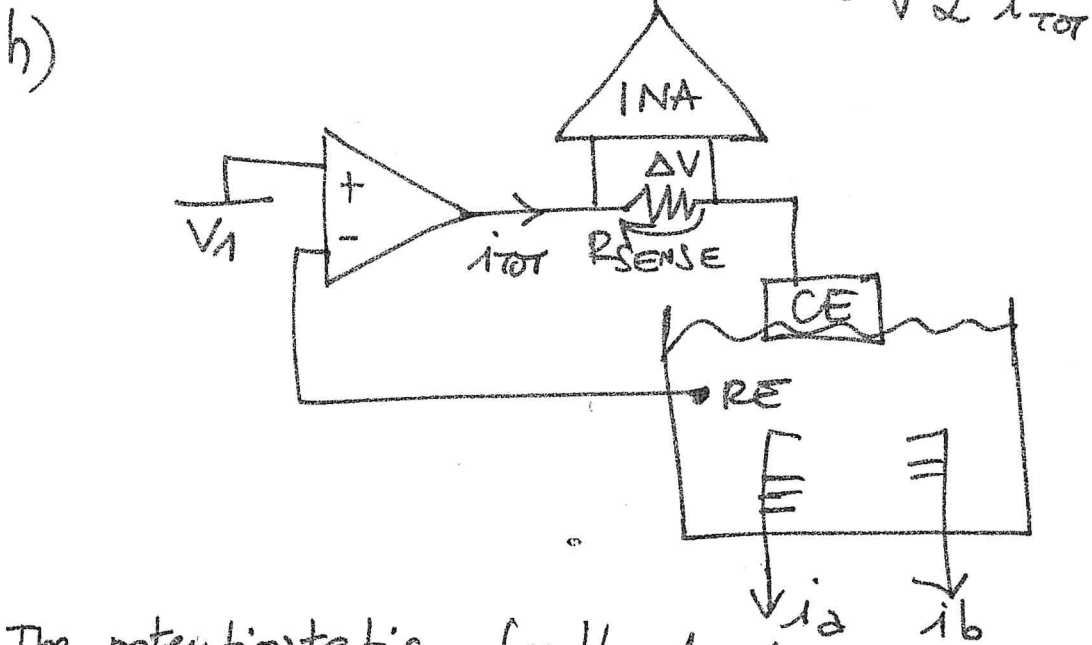
f) *fatto fino qui*

$$i_{REDOX} \propto \text{Area}$$

$$i_{REDOX} \propto \sqrt{\text{SCAN RATE}}$$

$$\text{AREA} \downarrow 2 \Rightarrow \text{SCAN RATE INCREASE by } 4$$

g) *Are* is the min. feature $5 \mu\text{m} \rightarrow$ accuracy better than (and G) if WE_a and WE_b DIFFERENT METALS, otherwise SINGLE MASK



The potentiostatic feedback loop compensates for the Ohmic drop ΔV . (amplified by the INA)

(COMMON MODE
check)